

Economics: Foundations and Models

ECON 101 - Macroeconomics

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Fall 2026

- 1.1 Three Key Economic Ideas
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- 1.3 Economic Models
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What Is Economics? (1 of 2)

- People make **choices** as they try to attain their goals.
- Choices are necessary because we live in a world of **scarcity**.
- **Scarcity**: a situation in which unlimited wants exceed the limited resources available to fulfill those wants.
- **Economics** is the study of the choices people make to attain their goals, given their scarce resources.

What Is Economics? (2 of 2)

- Economists study these choices using **economic models**:
 - **Economic model**: a simplified version of reality used to analyze real-world economic situations.
- Typical economics questions:
 - How are the prices of goods and services determined?
 - How do government spending and taxation affect the economy?
 - Why do governments control the prices of some goods?
 - Why are some countries wealthier than others?

Everyday Examples of Scarcity

- **Your time:** cannot work, study, and relax at the same time.
- **Government budgets:** more spending on health care may mean less on education or infrastructure.
- **Environmental trade-offs:** more production today may mean less clean air tomorrow.

Three Key Economic Ideas

- People are **rational**.
- People respond to **incentives**.
- Optimal decisions are made at the **margin**.

1. People Are Rational

- Economists generally assume that people are **rational**.
- **Rational**: people make choices based on what they believe will make them happy.
- They do not deliberately do things to make themselves worse off.
- Rational decision makers weigh **costs** and **benefits**.

Example: A \$5 Cup of Coffee

- You will not buy a \$5 cup of coffee unless you believe the benefit is worth at least:
 - the time and effort it takes you to earn \$5, **plus**
 - the value of any other item you could have bought with the \$5.
- **Hubbard-style interpretation:** Think in terms of **opportunity cost**:
 - If the best alternative use of \$5 is a bus pass or saving for tuition, you compare happiness from coffee vs. that alternative.

2. People Respond to Incentives

- As **costs** and **benefits** change, so do people's actions.
- Example: Do Canadian banks choose to get robbed?
 - Bullet-resistant shields cost \$20,000.
 - Typical loss due to an armed robbery is \$1,200.
 - Armed robbery is rare.
 - **Result:** Many banks choose not to install bullet-resistant shields.

- **Students:** Higher course weight or harder marking may increase study time.
- **Firms:** A carbon tax raises the cost of emissions, encouraging cleaner technologies.
- **Households:** Higher interest rates increase the reward to saving but raise borrowing costs.

3. Optimal Decisions Are Made at the Margin

- Most decisions involve doing a **little more** or a **little less** of something.
- Example: Should you watch an extra hour of TV or study instead?
- **Marginal cost (MC)**: the additional cost from a small increase in activity.
- **Marginal benefit (MB)**: the additional benefit from that small increase.
- Comparing MC and MB is called **marginal analysis**.

Worked Example: Study vs. TV

Suppose:

- Extra hour of TV: $MB = 8$ “happiness points”, $MC = 0$ points.
- Extra hour of study: increases exam grade by 2 percentage points, which you value at 12 points of happiness (less stress, better GPA).
- **Decision:** If you switch from TV to study:
 - $MB(\text{study}) = 12$, $MC(\text{giving up TV}) = 8$.
 - Net gain = 4 points $\Rightarrow$ better to study the extra hour.

1.2 Learning Objective

Discuss how an economy answers these questions:

- **What** goods and services will be produced?
- **How** will the goods and services be produced?
- **Who** will receive the goods and services produced?

1. What Goods and Services Will Be Produced?

- All societies must decide which goods and services to produce.
- Producing more of one thing means producing less of another.
- This is a **trade-off**, resulting from the scarcity of productive resources.
- **Opportunity cost**: the value of the best alternative given up.
- Example: The opportunity cost of increased funding for health care might be fewer university scholarships.

2. How Will the Goods Be Produced?

- A firm may have multiple methods for producing the same output.
- **Example 1 (Music production):**
 - Hire a great singer and use standard production techniques; or
 - Hire a mediocre singer and use Auto-Tune to correct inaccuracies.
- **Example 2 (Manufacturing labour):**
 - As labour costs rise, firms may substitute capital (machines) for labour; or
 - Move production to a country with cheaper labour.

3. Who Will Receive the Goods and Services?

- Who gets what is closely tied to **income distribution**.
- People with higher incomes can buy more goods and services.
- Governments can change the income distribution through:
 - **Taxes**
 - **Transfers** (e.g., EI, child benefits, social assistance)

Centrally Planned vs. Market vs. Mixed Economies

- **Centrally planned economy:** government decides how economic resources will be allocated.
- **Market economy:** decisions of households and firms interacting in markets allocate resources.
- **Mixed economy:** most decisions are market-based, but government plays a significant role.
- Canada today is a **mixed economy**:
 - Hospitals: centrally planned by provincial governments.
 - Fast food: organized by markets.

- Market economies tend to be more **efficient** than centrally planned economies.
- **Productive efficiency**: goods and services produced at the lowest possible cost.
- **Allocative efficiency**: production in accordance with consumer preferences:
 - Each good produced up to the point where last unit's **MB** equals its **MC**.
- **Equity**: the fair distribution of economic benefits.

- **Productive efficiency** comes from **competition** among firms.
- **Allocative efficiency** arises from **voluntary exchange**:
 - When buyer and seller both benefit from trade.
 - Transactions continue until no further mutual gains are possible.

- Markets may not result in fully efficient outcomes:
 - It takes time to discover the most efficient production methods.
 - Governments may interfere with market outcomes (taxes, controls).
 - Market outcomes can ignore those not directly involved:
 - **Externalities** (e.g., pollution).
- Economic efficiency is not the only objective; **equity** matters too.

- Economists develop models to analyze real-world issues.
- Building a model often follows these steps:
 - Decide on the assumptions.
 - Formulate a testable hypothesis.
 - Use economic data to test the hypothesis.
 - Revise the model if it fails to explain the data well.
 - Retain the revised model for similar questions in the future.

Important Features of Economic Models

- **Assumptions:** every model needs them to be tractable and useful.
- **Hypothesis testing:** good models generate testable predictions.
- **Economic variable:** something measurable that can take different values (e.g., income, unemployment rate).

The Scientific Nature of Economics

- Economists try to mimic the scientific method:
 - Develop models
 - Derive predictions
 - Compare with data
- Economics is a **social science**:
 - People may have strong views about what results “should” be.

Positive vs. Normative Analysis

- **Positive analysis:** based on facts and logic.
 - “If the minimum wage rises, some low-skill employment will fall.”
- **Normative analysis:** based on value judgments.
 - “The minimum wage should be higher to ensure a living standard.”
 - Words like “should” or “ought” often signal normative statements.
- Economics **tries** to focus on positive analysis, while recognizing normative debates.

Apply the Concept: Cap on House Prices (1 of 2)

- Consider a government **cap** on house prices.
- **Potential benefits:**
 - Lower home prices for buyers.
 - Lower housing costs free up income for other spending.
- **Potential costs:**
 - Fewer houses built.
 - Fewer jobs in construction and related sectors.

Apply the Concept: Cap on House Prices (2 of 2)

- Both benefits and costs are **positive** effects we can try to measure.
- How much each effect matters is a **normative** judgment.
- This example illustrates:
 - Trade-offs and opportunity cost.
 - Distinction between positive and normative analysis.

1.4 Learning Objective

Distinguish between microeconomics and macroeconomics.

- **Microeconomics:** the study of how households and firms make choices, how they interact in markets, and how governments attempt to influence their choices.
- **Macroeconomics:** the study of the economy as a whole, including inflation, unemployment, and economic growth.

Examples: Micro vs. Macro Questions

- **Micro:**

- Why did Netflix raise its subscription price?
- How will a tax on sugary drinks affect soft drink sales?

- **Macro:**

- Why does Canada experience recessions?
- What causes inflation to rise or fall?

1.5 Learning Objective

Define important economic terms.

- Economics has its own terms and jargon with specific, precise meanings.
- Sometimes these terms are used differently than in everyday language.

The Language of Economics (2 of 2)

- **Technology:** the processes a firm uses to turn inputs into outputs of goods and services.
- **Capital:** durable manufactured goods used to produce other goods and services.
- Pay close attention to terms defined in class and in the textbook:
 - Using terms correctly helps you think like an economist.
 - On exams, definitions are often tested directly or indirectly.

- Economics is about how individuals and societies make choices under **scarcity**.
- Three key ideas:
 - People are rational.
 - People respond to incentives.
 - Optimal decisions are made at the margin.
- Societies face three fundamental questions: what, how, and who.
- Market, centrally planned, and mixed economies differ in how they answer these questions.

Summary (continued)

- Economic models simplify reality to generate testable predictions.
- Economists distinguish between **positive** (what is) and **normative** (what ought to be) analysis.
- Microeconomics focuses on individual markets; macroeconomics looks at the economy as a whole.
- Mastering the **language of economics** (terms like scarcity, opportunity cost, technology, capital) is essential for success in this course.

Thank you!